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24478

B. Tech. 7th Semester (ME) (Common with Special Chance Examination – December, 2019

OPERATION RESEARCH

Paper : ME-405-F

Time : Three Hours] *[Maximum Marks : 100*

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt ***five*** questions in all, selecting ***one*** question from each Section. Question No. **1** is ***compulsory***.

1. (a) Explain standard form of linear programming problem. $5 \times 4 = 20$

(b) Explain North-West corner rule for finding initial solution for a transportation problem.

(c) Explain the term PERT.

(d) What is simulation ? Describe its advantages in solving the problems.

SECTION – A

2. (a) Define operation research. What is the role of operation research in decision making ? 10

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(b) Discuss the various phases in solving an operation research problem. 10

3. Maximize $z = 3x_1 - x_2$, subject to constraints :

$$2x_1 + x_2 \le 2,$$

$$x_1 + 3x_2 \ge 3,$$

$$x_2 \le 4,$$

$$x_1, x_2 \ge 0$$

SECTION – B

4. Find the optimum solution to the following transportation problem in which the cells contain the transportation cost in rupees. 20

	W_1	W_2	W_3	W_4	W_5	Available
F_1	7	6	4	5	9	40
F_2	8	5	6	7	8	30
F_3	6	8	9	6	5	20
F_4	5	7	7	8	6	10
Required	30	30	15	20	5	100

5. (a) Explain the primal dual relationships. 5

(b) Using dual simplex method, maximize $z = -3x_1 - x_2$, subject to constraints : 15

$$x_1 + x_2 \ge 1,$$

$$2x_1 + 3x_2 \ge 2,$$

$$x_1, x_2 \ge 0$$

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SECTION – C

6. (a) Discuss basic elements of waiting line situations. 10

(b) Describe the applications of queuing theory. 10

7. A project schedule has the following characteristics : 20

Activity	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-7	6-7
Time	2	8	10	6	3	3	7	5	2	8

(a) Construct the network.

(b) Compute earliest occurrence time and latest occurrence time for each event.

(c) Find the critical path.

(d) Determine total, free and independent floats.

SECTION – D

8. (a) Explain Monte Carlo method and its applications in industries. https://www.mdustudy.com 10

(b) Using mixed congruential method, generate a sequence of five three digit random numbers such that : 10

$x_{n+1} = (21x_n + 53)(\text{modulo } 100)$ and $x_0 = 46$.

9. (a) Describe the steps involved in the process of decision making. 10

(b) What is decision making under uncertainty ? Name commonly used criteria for solving problems under condition of uncertainty. Explain any *two* of them. 10

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